# CST8507: Natural Language Processing

# Assignment 1

## Part 1: Comparative Analysis of Text Representation

## Learning Objectives

- Apply classical machine learning classifiers to text data.
- Compare performance across different feature representations.

## Learning Resources

Lecture Slides, in class code and resources including Hybrid work (week 2 - 5).

## Overview

The goal of this part is to look for the best model that can classify Amazon reviews into five categories (1-5) using different textual features. The categories include business, entertainment, politics, and sports. The performance of each technique will be evaluated using accuracy and confusion matrix,

### Dataset:

Download Reviews.csv dataset from
https://www.kaggle.com/datasets/jagdishchavan/amazon-reviews

## If your computer cannot handle analyzing the entire dataset, you may randomly select a subset of the data, but it must include at least 10,000 records.

## Processing:

1. **Split the dataset** into a train set and a test set with a common ratio being 80% for the train set and 20% for the test set. Perform steps from 3

onwards on train dataset then test dataset. It is recommended to split the data into train and test sets before preprocessing and feature extraction. The reason for this is that preprocessing, and feature extraction can significantly change the distribution of the data, and it is important to ensure that the train and test sets are representative of the original data distribution.

2. **Data Preprocessing**: Perform necessary preprocessing steps on the collected data. **Explain your preprocessing steps in your report.**
3. **Feature Extraction**: Extract features from the preprocessed data using **n-gram(*choose n*), Tf-Idf and one of the embedding techniques (Word*2Vec, GloVe or FastText*).**
4. **Modeling:** Use the extracted feature vectors as input to **two different machine learning models of your choice**. Evaluate the models using 'Score' as the label.
5. **Model Evaluation**: The model should be evaluated on the test set using the accuracy metric. This will give an idea of how well the model is able to classify new unseen data.
6. **Tune the model's hyperparameters**: If the model does not perform well on the test set, the model's hyperparameters can be fine-tuned to improve performance.

## Include The Following Information in Your Report:

1. Explain how you preprocess the data.
2. Include the confusion matrix for all the models.
3. Your interpretation and discussion of the results.
4. Discuss how you Tune your model's hyperparameters to get the best accuracy results.
5. Fill in the best results for all the used methods:

| Feature Type | Model | Accuracy | F-score |
|---|---|---|---|
| Embedding | | | |

| | | | |
|---|---|---|---|
| n-gram | | | |
| TF-IDF | | | |

# Part 2: Error Analysis Across Feature Representations

The goal of this task is to interpret model behavior by analyzing misclassified text instances across different feature extraction techniques.

For each representation:

1. Extract all test instances where the predicted label differs from the true label.
2. Select **at least 10 misclassified examples per representation.**
3. identify and analyze:
    - Texts that are misclassified by **all representations**
    - Texts misclassified by **only simpler models** (BoW / TF–IDF)
    - Texts misclassified by **embedding-based models only**
4. Explain what linguistic or semantic features may have caused the error.

# Submission Instruction

Submit your Report and code **as a Jupyter Notebook with the running code**, you can do the following steps:

1. Open your Jupyter Notebook: You can open Jupyter Notebook either from the command line by typing "jupyter notebook" or from Anaconda Navigator.
2. Write your code: Write the code you want to submit in the cells of the Jupyter Notebook. **Make sure to run the cells so that the output is generated**.
3. Save the Notebook: Once you have written and run your code, save the Jupyter Notebook by clicking on File -> Save and Checkpoint or by pressing "Ctrl + S".

4. Export the Notebook: To export the Jupyter Notebook, go to File -> Download as -> Notebook (.ipynb). This will download the Notebook as a .ipynb file on your laptop.
5. Create a folder named "Assignment 1", Inside the " Assignment 1" folder, include:
    - "Report.doc"
    - "Assignemnt1_Code.ipynb "

Zip the Assignment 1" folder and Submit the zipped folder on Brightspace.

You can submit multiple times, with only the most recent submission (before the due date) graded. There will be mark deduction if you are not following the submission requirements.

# Grading Criteria

| **Criteria** | **Points** |
|---|---|
| Preprocessing | 10 |
| BoW Implementation | 10 |
| TF–IDF Implementation | 10 |
| Word Embeddings Implementation | 20 |
| Comparative Analysis | 25 |
| Error Analysis | 10 |
| Report and discussion | 15 |

# Due Date

Check Brightspace for due dates.